

DC-DC BOOST CONVERTER PROTOTYPE PWM CONTROL, THEORY & SIMULATION STUDY

For PV-Integrated Electric Vehicle (Scale-Down Prototype)

20W Solar Panel | 48V Battery Output | DC-DC Boost Converter Design

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1. Introduction

This document presents the theoretical and practical study of PWM (Pulse Width Modulation) control in a DC-DC Boost Converter designed for a PV-Integrated Electric Vehicle (EV) charging prototype. This report covers the operating principle of the boost converter, open-loop and closed-loop PWM control theory, voltage feedback using a voltage divider and Arduino-based duty cycle control..

2. Boost Converter Operating Principle

A boost converter is a DC-DC power converter that steps up an input voltage to a higher output voltage. It this through the controlled switching of a MOSFET, an inductor for energy storage, a diode for unidirectional current flow, and a capacitor for output voltage smoothing. The fundamental operating principle is governed by the relationship between the input voltage, output voltage, and duty cycle D.

2.1 Formula

The ideal voltage conversion ratio of a boost converter is given by:

$$V_{out} = V_{in} / (1 - D)$$

Where D is the duty cycle (ratio of MOSFET ON time to total switching period).

For this design:

$$\begin{aligned} D_{max} &= 1 - \frac{V_{in(min)}}{V_{out}} \\ D_{max} &= 1 - \frac{17.93}{48} \\ D_{max} &= 0.6265 \\ \mathbf{D} &= \mathbf{62.65\%} \end{aligned}$$

2.2 Switching Modes

When the MOSFET is ON (for time D.T), the diode is reverse biased and the inductor stores energy from the source. When the MOSFET is OFF (for time (1-D).T), the inductor releases its stored energy through the diode to the output capacitor and load. This energy transfer mechanism produces a boosted output voltage. The switching frequency in this design is 20 kHz (T = 50 µs per cycle).

3. Design Parameters

Parameter	Value
Solar Panel Power	20 W
Panel Voltage (V_{mp})	17.93 V
Open Circuit Voltage	21.4 V
Short Circuit Current	1.22 A
Max Power Current	1.11 A
Output Voltage	48 V
Switching Frequency	20,000 Hz
Duty Cycle	62.65%
Inductor	2.2 mH
Capacitor	47 μ F / 100V
Load Resistance	120 Ω / 25W
Input Current	1.11 A
Output Current	0.417 A

4. PWM Control Theory

Pulse Width Modulation (PWM) is the technique used to control the MOSFET switching in the boost converter. The Arduino generates a square wave signal at 20 kHz. The ratio of the ON time to the total period is the duty cycle D.

By controlling D, the output voltage is regulated according to the boost formula.

4.1 PWM Generation in the Circuit

In the PSCAD simulation and also in the physical prototype, PWM is generated by comparing a constant duty cycle reference signal with a high-frequency triangular wave using a comparator:

- The triangular wave oscillates between 0 and 1 at 20 kHz
- The duty cycle reference ($myduty = 0.6265$) is a flat constant signal
- When triangular wave < $myduty$: comparator output = HIGH (MOSFET ON)
- When triangular wave > $myduty$: comparator output = LOW (MOSFET OFF)
- The resulting square wave at the comparator output has $D = 62.65\%$ at 20 kHz

This is how real PWM integrated circuits (such as the SG3525 or TL494) function internally. They all contain an internal triangle oscillator performing this same comparison operation.

5. Open-Loop Control

In open-loop control, the duty cycle D is fixed at a calculated value (0.6265) whether input voltage or load changes.

The Arduino uses this value and there is no feedback from the output. Open-loop is a simple way to implement and function the converter for a fixed-input prototype demonstration.

5.1 Limitation of Open-Loop

If the input voltage changes due to varying solar irradiance, the output voltage will change proportionally since D is fixed. But in the open-loop when D is a fixed value, the converter cannot output the 48V output. This is a fundamental limitation of open-loop control and demonstrates the need for closed-loop feedback in real applications.

6. Closed-Loop Control with Voltage Feedback

In closed-loop control, the output voltage is continuously measured and fed back to the Arduino. The Arduino compares the measured voltage to the 48V target and automatically adjusts the duty cycle to correct any deviation. This ensures V_{out} stays at 48V regardless of changes in V_{in} or load.

6.1 Voltage Divider for Feedback

Since the Arduino analog pins can only measure 0 to 5V, a voltage divider is required to scale the 48V output down to a safe level. The divider is designed so that when $V_{out} = 48V$, the Arduino reads exactly 5V:

$$V_{read} = V_{out} \frac{R_2}{(R_1 + R_2)} = 5V$$

$$R_2 / (R_1 + R_2) = 5/48 = 0.1042$$

$$\text{Selected: } R_1 = 8.2 \text{ k}\Omega, R_2 = 1 \text{ k}\Omega$$

The voltage divider is connected directly across the output (48V rail to GND), with the midpoint (junction of R_1 and R_2) connected to Arduino analog pin A0. This scales 48V safely down to approximately 5V for the Arduino.

6.2 PI Controller in Arduino

The Arduino implements a Proportional-Integral (PI) control algorithm that runs every 10ms. The PI controller calculates an error (target - measured), accumulates the integral of the error over time, and adjusts the OCR1A register (which directly sets the duty cycle) to drive the error toward zero. Proportional action gives immediate response to error, while integral action eliminates steady-state error.

6.3 Comparison: Open-Loop vs Closed-Loop

Feature	Open-Loop	Closed-Loop
Complexity	Simple	Higher than open-loop
V_{out} stable if V_{in} changes?	No	Yes
Feedback required?	No	Yes (voltage divider + Arduino)
Real-world applicability	Limited	Industry standard